

Chemical Reactions and Equations

Science • Chapter 1 • Chapter Summary & Study Guide for Daksh

Everything around you — rusting of iron, ripening of fruits, digestion of food, burning of fuel — involves a **chemical reaction**. A chemical reaction is a process in which one or more substances (called **reactants**) are converted into one or more new substances (called **products**). The new substances have completely different properties from the original ones. This chapter teaches you how to recognise such changes, how to write them as equations, and how to classify them by type.

How do you know a chemical reaction has happened?

We look for tell-tale **observations**: a change in state, a change in colour, evolution of a gas, change in temperature, or formation of a precipitate. Any one of these is usually a clue that bonds are being broken and formed — the very essence of a chemical reaction.

- Magnesium ribbon burns with a **dazzling white flame** in air to give a white powder (magnesium oxide).
- Lead nitrate solution + Potassium iodide solution → bright yellow lead iodide **precipitate**.
- Zinc granules + dilute HCl → fizzes; **hydrogen gas** bubbles out and the test tube feels warm.

Writing chemical equations & balancing them

A **chemical equation** uses formulae instead of words. The skeletal (unbalanced) form is fine to start with, but a proper equation must obey the **Law of Conservation of Mass** — atoms are neither created nor destroyed in an ordinary chemical reaction. So the number of atoms of every element on the left (reactant side) must equal the number on the right (product side). We achieve this by adjusting **coefficients**, never by changing formulae.

Skeletal :	Fe	+	H ₂ O	->	Fe ₃ O ₄	+	H ₂	(NOT balanced)
Balanced :	3Fe	+	4H ₂ O	->	Fe ₃ O ₄	+	4H ₂	(balanced)

Conditions can also be written above the arrow:

-> heat (Delta), -> catalyst, -> electricity, -> sunlight

Types of chemical reactions

Chemists classify reactions into a few broad **types** so we can predict what will happen in similar situations.

- **Combination**: two or more reactants combine to form a single product. e.g. $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ (quick lime + water gives slaked lime, with a lot of heat — an **exothermic** reaction).
- **Decomposition**: a single reactant breaks down into two or more products. Energy is needed — supplied as heat (*thermal*), light (*photolytic*) or electricity (*electrolytic*). e.g. $2\text{FeSO}_4 \xrightarrow{(\Delta)} \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$; $2\text{AgCl} \xrightarrow{(\text{sunlight})} 2\text{Ag} + \text{Cl}_2$; $2\text{H}_2\text{O} \xrightarrow{(\text{electricity})} 2\text{H}_2 + \text{O}_2$.
- **Displacement**: a more reactive metal pushes out a less reactive metal from its salt solution. e.g. $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ (the blue colour fades, brown copper is deposited on the iron nail).
- **Double displacement**: two compounds exchange their ions; often a precipitate or gas forms. e.g. $\text{Pb(NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 \downarrow (\text{yellow}) + 2\text{KNO}_3$.
- **Oxidation & Reduction (Redox)**: oxidation = gain of oxygen / loss of hydrogen / loss of electrons; reduction = the opposite. Both happen together. e.g. $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$. CuO loses O (reduced); H₂ gains O (oxidised).

Reactions are also classified as **exothermic** (release energy — respiration, combustion of LPG, mixing slaked lime in water) or **endothermic** (absorb energy — decomposition of CaCO_3 on heating, photosynthesis).

Effects of oxidation in everyday life

Two oxidation effects meet you every day at home and outside:

- **Corrosion** — slow eating up of a metal by air/moisture/chemicals. Iron rusts (reddish-brown $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$); silver tarnishes black (Ag_2S); copper develops a green coat (basic copper carbonate). Rusting is prevented by painting, oiling, greasing, galvanisation (zinc coating), chromium plating, alloying (e.g. stainless steel).

- **Rancidity** — oils and fats in food get oxidised on long exposure to air, giving an unpleasant smell and taste. We delay this by adding *antioxidants*, by flushing food packets with **nitrogen** (e.g. chip packets), by refrigeration, and by storing food in airtight containers.

Quick recap of skills you should master

- **Balance** a skeletal equation by the hit-and-trial method (start with the most complex molecule).
- Add **state symbols** (s/l/g/aq) and conditions over arrows.
- **Identify** the type of a given reaction from its equation.
- Spot **oxidising** and **reducing** agents in a redox reaction.
- Recognise an exothermic vs endothermic reaction from a description.

★ MUST-REMEMBER POINTS

- Balanced equation → atoms of every element are equal on both sides (Law of Conservation of Mass).
- Five reaction types: **Combination, Decomposition, Displacement, Double displacement, Redox**.
- Endothermic absorbs heat; Exothermic releases heat. Respiration is exothermic; photosynthesis is endothermic.
- Activity series matters in displacement: more reactive metal displaces less reactive one.
- Rancidity is delayed by antioxidants and inert-gas (N_2) packing.
- Galvanisation = coating with zinc to prevent rusting.

— End of Summary —